

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau
Practical – IX

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Cleaning and Structuring Data for Analysis

A) Identify and handle missing or inconsistent data.

Steps:-

1. Open **Tableau Desktop**, load the given dataset and go to **Sheet 1**.
2. Right-click in the **Data Pane** → **Create Calculated Field**, name it **Physics_Clean**, and enter: **FLOAT(REPLACE(STR([Physics]), '%', ''))**; similarly create **Attendance_Clean** using **FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))**.
3. Create calculated fields **Math_Clean** and **Chemistry_Clean** using the formulas: **IFNULL([Math], 0)** and **IFNULL([Chemistry], 0)**.
4. Edit **Physics_Clean** and use: **IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)** to handle missing Physics values.
5. Click **OK**, verify the cleaned fields in the Data Pane, and use the cleaned data for analysis.

Output:-

The screenshot displays the Tableau Desktop interface. On the left, the 'Data Pane' shows the 'Sem1_Scores.csv' dataset. The main view shows a table of data with columns: Student ID, Full Name, Dept Info, Math, Physics, Chemistry, Programming, and Attendance. The data rows show scores for various students, with some missing values (null) in the Math and Chemistry columns.

Below the data table, the 'Create Calculated Field' dialog is shown, listing the following fields:

- Physics_clean**: `FLOAT(REPLACE(STR([Physics]), '%', ''))`
- Attendance_clean**: `FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`
- Math_clean**: `IFNULL([Math], 0)`
- Chemistry_clean**: `IFNULL([Chemistry], 0)`

The 'Measure Values' section shows the following fields:

- Physics_clean**
- Attendance_clean**
- Math_clean**
- Chemistry_clean**

B) Convert wide data into tall format.

Steps:-

1. Open the **Data Source** tab in Tableau and select the required dataset.
2. Hold **Ctrl** and select **Math**, **Physics**, **Chemistry**, and **Programming** columns.
3. Right-click on the selected columns and choose **Pivot** to convert the wide data into tall format.
4. Rename **Pivot Field Names** as **Subject** and **Pivot Field Values** as **Score**.

5. Create a calculated field **Clean_Score** using **IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)** and verify the converted data.

Output:-

#	Math	Sem1_Scores.csv	Math
#	Physics	Sem1_Scores.csv	Physics
#	Chemistry	Sem1_Scores.csv	Chemistry
#	Programming	Sem1_Scores.csv	Programming

#	Math	
#	Physics	Create Calculated Field...
#	Chemistry	Pivot
#	Programming	Merge Mismatched Fields

Sem1
 Name
 Description
 No description
 Fields
 Type
 Abc
 Abc
 Abc

Number (decimal)
 Number (whole)
 Date & Time
 Date
 ✓ String
 Spatial
 Boolean
 ✓ Default
 Geographic Role
 Image Role

Abc	Subject
#	Score

clean_score

IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)

C) Perform union of multiple files.

Steps:

1. Open the Data Source tab and clear the existing canvas if required.
2. Drag New Union onto the canvas and add **Sem1_Scores.csv** and **Sem2_Scores.csv**.
3. Check the data grid at the bottom and verify that the combined dataset contains 20 rows.
4. Scroll to the Table Name column and verify that the first 10 rows belong to **Sem1_Scores.csv** and the remaining 10 rows belong to **Sem2_Scores.csv**.
5. Verify the merged dataset and click Sheet 1 to use the Union data for analysis.

Output:-

Union
 Specific (manual)
 Wildcard (automatic)

Connection: Sem1_Scores
 Sem1_Scores.csv
 Sem2_Scores.csv

Tables in union: 2
 Apply
 OK

Filters
 Sheet 1
 Semester

Marks
 Automatic
 Color
 Size
 Text
 Detail
 Tooltip

IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
 ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
 ELSE "Other"
 END

The calculation is valid.
 Apply
 OK

D) Execute cross-database joins.

Steps:

1. Stay on the Data Source canvas and keep the Union table on the canvas.
2. Drag **Department_Master.csv** to the right of the Union table and connect **Dept_Info** with **Dept_Code** using the **=** operator.
3. Drag **Student_Activities.csv** onto the canvas and connect **Student_ID** from the Union table with **Student_ID** from Student_Activities using **=**.
4. Click Sheet 1 and check the Data Pane for fields from Union, Department_Master, and Student_Activities.
5. Verify that the related fields are available and use the combined data for further analysis and visualization.

Output:-

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Sem1_Scores' as a text file. Below it, the 'Files' pane shows a list of CSV files including '2026_team_summaries.csv', 'comprehensiv..._nds_data.csv', 'Department_Master.csv', 'Eula.txt', 'Sample - Superstore.csv', 'Sem1_Scores.csv', 'Sem2_Scores.csv', 'startup_funding.csv', 'store_sales_data (2).csv', 'student-mat.csv', 'Student_Activities.csv', 'supermarket_sales.csv', 'superstore - superstore.csv', 'supply_chain_data (1).csv', and 'supply_chain_data.csv'. The main workspace shows a 'Union' join of 'Department_Master.csv' and 'Student_Activities.csv'. The 'Union' is named 'Student_A...'. The 'Fields' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Columns' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Rows' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Marks' shelf contains 'Student_Activities.csv'. The 'Marks' card is set to 'Club Membership'. The 'Marks' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Marks' card is set to 'Club Membership'. The 'Marks' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Marks' card is set to 'Club Membership'.

E) Restructure data for better visualization.**Steps:**

1. Open Sheet 1, create a calculated field named **Total_Score**, and enter: **[Math_Clean] + [Physics_Clean] + [Chemistry_Clean] + [Programming]**.
2. Create **Overall_Percentage** using: **[Total_Score] / 400.0**, then format it as Percentage with 1 decimal place.
3. Create **Performance_Band** using: **IF [Overall_Percentage] >= 0.85 THEN "Distinction" ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class" ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class" ELSE "Needs Academic Support" END.**
4. Create **Attendance_Status** using: **IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)" ELSE "Shortage Warning (<75%)" END.**
5. Build the scorecard by placing **Student_ID** and **Full_Name** on Rows, **Semester** on Columns, **Overall_Percentage** on Text, and **Performance_Band** on Color.

Output:-

The screenshot shows the Tableau Desktop interface. On the left, the 'Connections' pane lists 'Sem1_Scores' as a text file. Below it, the 'Files' pane shows a list of CSV files including '2026_team_summaries.csv', 'comprehensiv..._nds_data.csv', 'Department_Master.csv', 'Eula.txt', 'Sample - Superstore.csv', 'Sem1_Scores.csv', 'Sem2_Scores.csv', 'startup_funding.csv', 'store_sales_data (2).csv', 'student-mat.csv', 'Student_Activities.csv', 'supermarket_sales.csv', 'superstore - superstore.csv', 'supply_chain_data (1).csv', and 'supply_chain_data.csv'. The main workspace shows a 'Union' join of 'Department_Master.csv' and 'Student_Activities.csv'. The 'Union' is named 'Student_A...'. The 'Fields' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Columns' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Rows' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Marks' shelf contains 'Student_Activities.csv'. The 'Marks' card is set to 'Club Membership'. The 'Marks' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Marks' card is set to 'Club Membership'. The 'Marks' shelf contains 'Student_ID' and 'Student_Activities.csv'. The 'Marks' card is set to 'Club Membership'.

Performance_Band

```

IF [Overall_percentage] >= 0.85 THEN "Distinction"
ELSEIF [Overall_percentage] >= 0.75 THEN "First Class"
ELSEIF [Overall_percentage] >= 0.60 THEN "Second Class"
ELSE "Needs Academic Support"
END

```

The calculation is valid.

Apply OK

Attendance_Status

```

IF INT([Attendance]) >= 75 THEN
  "Eligible (Good Standing)"
ELSE
  "Shortage Warning (<75)"
END

```

The calculation is valid.

Apply OK

Data Analytics

Sem1_Scores

Tables

- Department_Master.c...
 - Dept Name
 - Dept Code
 - HOD Name
 - Dept Target Average
 - Total Dept Capacity
 - Department_Master...
- Student_Activities.csv
 - Club Membership
 - Student ID (Student...
 - Certifications
 - Projects Completed
 - Student_Activities.c...
- Union
 - Attendance
 - Attendance_Status
 - Dept Info

Filters

Marks

Color Size Text

Detail Tooltip

Performance...

ATTR(Attendance...

SUM(Overall...

Sheet 1

Student_ID	Full_Name	Semester	
		Semeste r1	Semeste r2
S101	Aarav Sharma	67.2%	69.2%
S102	Priya Verma		62.4%
S103	Rohan Gupta		63.2%
S104	Sneha Patel	69.7%	72.0%
S105	Aditya Rao		43.1%
S106	Ananya Singh	66.5%	68.5%
S107	Rahul Nair		58.7%
S108	Kavya Reddy	71.5%	72.7%
S109	Vikram Joshi	49.4%	51.9%
S110	Pooja Iyer	63.0%	65.5%

F) Handle different levels of detail in datasets

Steps:

1. Create **Dept_Average_Percentage_LOD** using: **{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }**.
2. Create **Student_Overall_Career_Average** using: **{ FIXED [Student_ID] : AVG([Overall_Percentage]) }**.
3. Create **Variance_From_Dept_Average** using: **[Overall_Percentage] - [Dept_Average_Percentage_LOD]**.
4. Create **Benchmark_Status** using: **IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark" ELSE "Below Department Benchmark" END**.
5. Create the LOD visualization by placing **Dept_Info**, **Student_ID**, and **Full_Name** on Rows, **Semester** on Columns, **Overall_Percentage** on Text, **Dept_Average_Percentage_LOD** on Tooltip, and **Benchmark_Status** on Color.

Output:-

Default Number Format [Dept_Average_Percentage_LOD]

Automatic

Number (Standard)

Number (Custom)

Currency (Standard)

Currency (Custom)

Scientific

Percentage

Custom

Percentage

Decimal places:

0

Clear OK Cancel

Variance_From_Dept_Average

```

[Overall_percentage] - [Dept_Average_Percentage_LOD]

```

The calculation is valid.

Apply OK

Benchmark_Status

```

IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"
ELSE "Below Department Benchmark"
END

```

The calculation is valid.

Apply OK

Dept_Average_Percentage_LOD

```

{ FIXED [Dept_Info] : AVG([Overall_percentage]) }

```

The calculation is valid.

Apply OK

Pages		Columns		Semester	
Filters		Rows		Dept_Info	
Marks		Student_ID		Full_Name	
Automatic		Color		Size	
Detail		Tooltip		Benchmark_St...	
SUM(Dept_Av...		SUM(Overall_p...			

Sheet 2		Semester	
Dept_Info	Student_ID	Full_Name	Semeste r 1
CSE-01	S101	Aarav Sharma	67.2%
	S103	Rohan Gupta	63.2%
	S106	Ananya Singh	66.5%
	S109	Vikram Joshi	49.4%
ECE-02	S102	Priya Verma	62.4%
	S107	Rahul Nair	58.7%
	S110	Pooja Iyer	63.0%
IT-03	S104	Sneha Patel	69.7%
	S108	Kavya Reddy	71.5%
MECH-04	S105	Aditya Rao	43.1%

G) Apply advanced data cleaning techniques.

Steps:

1. Create a calculated field named **Full_Name_Clean** and apply the name-cleaning formula using **TRIM**, **UPPER**, **LOWER**, **LEFT**, **MID**, **FIND**, and **LEN** functions.
2. Create **First_Name** using: **TRIM(SPLIT([Full_Name], " ", 1))** and create **Last_Name** using: **TRIM(SPLIT([Full_Name], " ", -1))**.
3. Create **Dept_Branch** using: **SPLIT([Dept_Info], "-", 1)** and **Section_ID** using: **SPLIT([Dept_Info], "-", 2)**.
4. Create **Student_Display_Label** using: **[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"**.
5. Click OK, verify all calculated fields in the Data Pane, and use the cleaned fields for visualization.

Output:-

Full_Name_Clean

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") -
1)) +
" " +
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1
1)) +
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2
LEN(TRIM([Full_Name]))))
```

The calculation is valid.

Apply OK

First_Name

```
TRIM(SPLIT([Full_Name], " ", 1))
```

The calculation is valid.

Apply OK

Last_Name

```
TRIM(SPLIT([Full_Name], " ", -1))
```

The calculation is valid.

Apply OK

DEpt_Branch

```
SPLIT([Dept_Info], "-", 1)
```

The calculation is valid.

Apply OK

Sheet 1

Section_ID

```
SPLIT([Dept_Info], "-", 2)
```

The calculation is valid.

Apply OK

Student_Display_Label

```
_ID] + " - " + [Full_Name_Clean] + " (" + [DEpt_Branch] + ") "
```

The calculation is valid.

Apply OK

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Data Analytics

Sem1_Scores

Search

Tables

Union

- Attendance
- Attendance_Status
- Benchmark_Status
- DEpt_Branch
- Dept_Info
- First_Name
- Full_Name
- Full_Name_Clean
- Last_Name
- Performance_Band
- Section_ID
- Semester
- Student_Display_La...
- Student_ID
- Table Name
- Chemistry
- Dept Average Perc...

Filters

DEpt_Branch

Section_ID

Marks

Automatic

Color Size Text

Detail Tooltip

Performance_...

SUM(Overall_...

Columns

Semester

Rows

Student_Display_Lab...

Sheet 1

Student_Display_Label	Semester	
	Semeste r 1	Semeste r 2
S101 - Aarav Sharma (CSE)	67.2%	69.2%
S102 - Priya Verma (ECE)		62.4%
S103 - Rohan Gupta (CSE)		63.2%
S106 - Ananya Singh (CSE)	66.5%	68.5%
S107 - Rahul Nair (ECE)		58.7%
S109 - Vikram Joshi (CSE)	49.4%	51.9%
S110 - Pooja Iyer (ECE)	63.0%	65.5%